



Dot Product of Vectors

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Learning Objectives

- Compute the dot product of two vectors using the formula $u \cdot v = ac + bd$
- Apply scalar multiplication in combination with the dot product
- Use properties of the dot product to simplify expressions
- Evaluate compound dot product expressions involving three vectors

Evaluate each dot product expression and show all work in the space provided.

1. Find the dot product of the vectors.

$$(4, 5) \cdot (2, 3)$$

Answer: _____

2. Find the dot product of the vectors.

$$(1, 7) \cdot (6, 2)$$

Answer: _____

3. Find the dot product of the vectors.

$$(-3, 4) \cdot (5, -2)$$

Answer: _____

4. Let $u=(2,3)$ and $v=(4,1)$. Find $u \cdot 2v$.

$$u \cdot 2v$$

Answer: _____

5. Let $u=(1,2)$, $v=(3,4)$, and $w=(5,6)$. Find $(u \cdot v)w$.

$$(u \cdot v)w$$

Answer: _____

6. Let $u=(2,-1)$ and $v=(3,5)$. Find $u \cdot 3v$.

$$u \cdot 3v$$

Answer: _____

7. Let $u=(2,1)$, $v=(1,3)$, and $w=(4,-2)$. Find $(u \cdot v)w$.

$$(u \cdot v)w$$

Answer: _____



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13-linear-algebra-vector-dot-product Worksheet

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8. Find the dot product of the vectors.

$$(0, 9) \cdot (7, -4)$$

Answer: _____

9. Let $u=(5,-2)$ and $v=(-1,4)$. Find $(2u) \cdot v$.

$$(2u) \cdot v$$

Answer: _____

10. Let $u=(3,2)$, $v=(2,-1)$, and $w=(1,4)$. Find $(u \cdot v)w$.

$$(u \cdot v) w$$

Answer: _____





This worksheet covers the dot product of two-dimensional vectors, including the basic formula $u \cdot v = ac + bd$, scalar multiples with the dot product such as $u \cdot (kv)$, and compound expressions like $(u \cdot v)w$ that combine scalar and vector results.

Solutions

1. Find the dot product of the vectors.

$$(4, 5) \cdot (2, 3)$$

- Multiply the first components: 4 times 2 equals 8.
- Multiply the second components: 5 times 3 equals 15.
- Add the two products: 8 plus 15 equals 23.

Answer: 23

2. Find the dot product of the vectors.

$$(1, 7) \cdot (6, 2)$$

- Multiply the first components: 1 times 6 equals 6.
- Multiply the second components: 7 times 2 equals 14.
- Add the two products: 6 plus 14 equals 20.

Answer: 20

3. Find the dot product of the vectors.

$$(-3, 4) \cdot (5, -2)$$

- Multiply the first components: negative 3 times 5 equals negative 15.
- Multiply the second components: 4 times negative 2 equals negative 8.
- Add the two products: negative 15 plus negative 8 equals negative 23.

Answer: -23

4. Let $u=(2,3)$ and $v=(4,1)$. Find $u \cdot 2v$.

$$u \cdot 2v$$

- Multiply vector v by 2 to get the vector with components 8 and 2.
- Take the dot product of u and the new vector: 2 times 8 equals 16 and 3 times 2 equals 6.
- Add the products: 16 plus 6 equals 22.

Answer: 22

5. Let $u=(1,2)$, $v=(3,4)$, and $w=(5,6)$. Find $(u \cdot v)w$.

$$(u \cdot v)w$$

- Find $u \cdot v$ by multiplying components: 1 times 3 equals 3 and 2 times 4 equals 8, so $u \cdot v$ equals 11.
- Multiply the scalar 11 by vector w : 11 times 5 equals 55 and 11 times 6 equals 66.
- The result is the vector with components 55 and 66.

Answer: (55, 66)



6. Let $u=(2,-1)$ and $v=(3,5)$. Find $u \cdot 3v$.

$$u \cdot 3v$$

→ Multiply vector v by 3 to get the vector with components 9 and 15.

→ Take the dot product with u : 2 times 9 equals 18 and negative 1 times 15 equals negative 15.

→ Add the products: 18 plus negative 15 equals 3.

Answer: 3

7. Let $u=(2,1)$, $v=(1,3)$, and $w=(4,-2)$. Find $(u \cdot v)w$.

$$(u \cdot v)w$$

→ Find $u \cdot v$: 2 times 1 equals 2 and 1 times 3 equals 3, so $u \cdot v$ equals 5.

→ Multiply the scalar 5 by w : 5 times 4 equals 20 and 5 times negative 2 equals negative 10.

→ The result is the vector with components 20 and negative 10.

Answer: (20, - 10)

8. Find the dot product of the vectors.

$$(0,9) \cdot (7, -4)$$

→ Multiply the first components: 0 times 7 equals 0.

→ Multiply the second components: 9 times negative 4 equals negative 36.

→ Add the products: 0 plus negative 36 equals negative 36.

Answer: -36

9. Let $u=(5,-2)$ and $v=(-1,4)$. Find $(2u) \cdot v$.

$$(2u) \cdot v$$

→ Multiply vector u by 2 to get the vector with components 10 and negative 4.

→ Take the dot product with v : 10 times negative 1 equals negative 10 and negative 4 times 4 equals negative 16.

→ Add the products: negative 10 plus negative 16 equals negative 26.

Answer: -26

10. Let $u=(3,2)$, $v=(2,-1)$, and $w=(1,4)$. Find $(u \cdot v)w$.

$$(u \cdot v)w$$

→ Find $u \cdot v$: 3 times 2 equals 6 and 2 times negative 1 equals negative 2, so $u \cdot v$ equals 4.

→ Multiply the scalar 4 by w : 4 times 1 equals 4 and 4 times 4 equals 16.

→ The result is the vector with components 4 and 16.

Answer: (4, 16)

